

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

09

8 answers · Rationale-rich · Teach from doc

Q1**B**

B. The estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice B is correct. On the graph shown, the y -axis represents estimated population, in thousands. The graph shows that when $x = 0$, the y -coordinate is 6. Therefore, the estimated population at $x = 0$ is 6 thousand. The graph also shows that when $x = 1$, the y -coordinate is 9. Therefore, the estimated population at $x = 1$ is 9 thousand. Dividing 9 thousand by 6 thousand yields 1.5; therefore, 9 thousand is 1.5 times 6 thousand. It follows that the estimated population at $x = 1$ is 1.5 times the estimated population at $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

D

D.

x	-1	0	1
$q(x)$	16	32	64

Choice D is correct. Substituting -1 for x in the given function yields $q(-1) = 322^{-1}$, which is equivalent to $q(-1) = 32^{-1}_2$, or $q(-1) = 16$. Therefore, when $x = -1$, the corresponding value of qx for function q is 16. Substituting 0 for x in the given function yields $q(0) = 322^0$, which is equivalent to $q(0) = 32^1$, or $q(0) = 32$. Therefore, when $x = 0$, the corresponding value of qx for function q is 32. Substituting 1 for x in the given function yields $q(1) = 322^1$, which is equivalent to $q(1) = 32^2$, or $q(1) = 64$. Therefore, when $x = 1$, the corresponding value of qx for function q is 64. Of the choices given, only the table in choice D gives these three values of x and their corresponding values of qx for function q .

Choice A is incorrect. This table gives three values of x and their corresponding values of qx for the function $qx = 322x$.

Choice B is incorrect. This table gives three values of x and their corresponding values of qx for the function $qx = 232^x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3

D

D. $hx = 1.231.12^x$

Choice D is correct. The table shows an increasing exponential relationship between the number of years, x , since Hana started training in pole vault and the estimated height hx , in meters, of her best pole vault for that year. The relationship can be written as $hx = Ca^x$, where C and a are positive constants. It's given that when $x = 0$, $hx = 1.23$. Substituting 0 for x and 1.23 for hx in $hx = Ca^x$ yields $1.23 = Ca^0$, or $1.23 = C$. Substituting 1.23 for C in $hx = Ca^x$ yields $hx = 1.23a^x$. It's also given that when $x = 2$, $hx = 1.54$. Substituting 2 for x and 1.54 for hx in $hx = 1.23a^x$ yields $1.54 = 1.23a^2$. Dividing each side of this equation by 1.23 yields $\frac{1.54}{1.23} = \frac{1.23a^2}{1.23}$, or a^2 is approximately equal to 1.252. Since a is positive, a is approximately equal to $\sqrt{1.252}$, or 1.12. Substituting 1.12 for a in $hx = 1.23a^x$ yields $hx = 1.231.12^x$.

Choice A is incorrect. When $x = 0$, the value of hx in this function is equal to 1.12 rather than 1.23, and it is decreasing rather than increasing.

Choice B is incorrect. When $x = 0$, the value of hx in this function is equal to 1.12 rather than 1.23.

Choice C is incorrect. This function is decreasing rather than increasing.

Q4

A

A. 3

Choice A is correct. It's given that an object's kinetic energy, in joules, is equal to the product of one-half the object's mass, in kilograms, and the square of the object's speed, in meters per second. This relationship can be represented by the equation $K = \frac{1}{2}mv^2$, where K is the kinetic energy, m is the mass, and v is the speed. Substituting a mass of 4 kilograms for m and a kinetic energy of 18 joules for K results in the equation $18 = \left(\frac{1}{2}\right)(4)v^2$, or $18 = 2v^2$. Dividing both sides of this equation by 2 yields $9 = v^2$. Taking the square root of both sides yields $v = -3$ and $v = 3$. Since speed can't be expressed as a negative number, the speed of the object is 3 meters per second.

Choice B is incorrect and may result from computation errors. Choice C is incorrect. This is the value of v^2 rather than v. Choice D is incorrect. This is the value of $4v^2$ rather than v.

Q5**B**

B. $\left(-\frac{2}{3}, 0\right)$

Choice B is correct. The graph of the function f in the xy -plane has x -intercepts at the points (x, y) , where $y = f(x) = 0$. Substituting 0 for $f(x)$ in the given equation yields $0 = (x+4)(x-1)(2x-3)$. By the zero product property, if $0 = (x+4)(x-1)(2x-3)$, then $x+4=0$, $x-1=0$, or $2x-3=0$. Solving each of these linear equations for x , it follows that $x = -4$, $x = 1$, and $x = \frac{3}{2}$, respectively. This means that the graph of the function f in the xy -plane has three x -intercepts: $(-4, 0)$, $(1, 0)$, and $\left(\frac{3}{2}, 0\right)$. Therefore, $\left(-\frac{2}{3}, 0\right)$ isn't an x -intercept of the graph of the function f .

Alternate approach: Substitution may be used. Since by definition an x -intercept of any graph is a point in the form $(k, 0)$ where k is a constant, and since all points in the options are in this form, it need only be checked whether the points in the options lie on the graph of the function f . Substituting $-\frac{2}{3}$ for x and 0 for $f(x)$ in the given equation yields

$$0 = \left(-\frac{2}{3} + 4\right)\left(-\frac{2}{3} - 1\right)\left(2\left(-\frac{2}{3}\right) - 3\right), \text{ or } 0 = \frac{650}{27}.$$

Therefore, the point $\left(-\frac{2}{3}, 0\right)$ doesn't lie on the graph of the function f and can't be an x -intercept of the graph.

Choices A, C, and D are incorrect because each of these points is an x -intercept of the graph of the function f in the xy -plane. By definition, an x -intercept is a point on the graph of the form $(k, 0)$, where k is a constant. Substituting -4 for x and 0 for $f(x)$ in the given equation yields $0 = (-4+4)(-4-1)(2(-4)-3)$, or $0 = 0$. Since this is a true statement, the point $(-4, 0)$ lies on the graph of the function f and is an x -intercept of the graph. Performing similar substitution using the points $(1, 0)$ and $\left(\frac{3}{2}, 0\right)$ also yields the true statement $0 = 0$, illustrating that these points also lie on the graph of the function f and are x -intercepts of the graph.

Q6**B****B. 18**

Choice B is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b represents the length of the base of the triangle and h represents its height. It's given that the area of a triangle is 270 square centimeters and that the length of the base of this triangle is 12 centimeters greater than the height of the triangle. Let x represent the height, in centimeters, of the triangle. It follows that the length of the base of the triangle can be expressed as $x + 12$. Substituting 270 for A , x for h , and $x + 12$ for b in the formula $A = \frac{1}{2}bh$ yields $270 = \frac{1}{2}(x + 12)(x)$, or $270 = \frac{1}{2}x(x + 12)$. Multiplying both sides of this equation by 2 yields $540 = x(x + 12)$. Applying the distributive property on the right-hand side of this equation yields $540 = x^2 + 12x$. Subtracting 540 from both sides of this equation yields $0 = x^2 + 12x - 540$. In factored form, this equation is equivalent to $(x + 30)(x - 18) = 0$. Applying the zero product property, it follows that $x + 30 = 0$ or $x - 18 = 0$. Subtracting 30 from both sides of the equation $x + 30 = 0$ yields $x = -30$. Adding 18 to both sides of the equation $x - 18 = 0$ yields $x = 18$. Since x represents the height of the triangle, it must be positive. Therefore, the height, in centimeters, of the triangle is 18.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**A****A.** 9

Choice A is correct. It's given that the volleyball court is rectangular and has an area of 162 square meters. The formula for the area of a rectangle is $A = \ell \cdot w$, where A is the area, ℓ is the length, and w is the width of the rectangle. It's also given that the length of the volleyball court is twice the width, thus $\ell = 2w$. Substituting the given value into the formula for the area of a rectangle and using the relationship between length and width for this rectangle yields $162 = (2w)(w)$. This equation can be rewritten as $162 = 2w^2$. Dividing both sides of this equation by 2 yields $81 = w^2$. Taking the square root of both sides of this equation yields $\pm 9 = w$. Since the width of a rectangle is a positive number, the width of the volleyball court is 9 meters.

Choice B is incorrect because this is the length of the rectangle. Choice C is incorrect because this is the result of using 162 as the perimeter rather than the area. Choice D is incorrect because this is the result of calculating w in the equation $162 = 2w + w$ instead of $162 = (2w)(w)$.

Q8**D****D.** $gx = \frac{1}{25}^x$

Choice D is correct. It's given that function g is exponential. Therefore, an equation defining g can be written in the form $gx = ab^x$, where a and b are constants. The table shows that when $x = 0$, $gx = 1$. Substituting 0 for x and 1 for gx in the equation $gx = ab^x$ yields $1 = ab^0$, which is equivalent to $1 = a$. Substituting 1 for a in the equation $gx = ab^x$ yields $gx = b^x$. The table also shows that when $x = 1$, $gx = \frac{1}{25}$. Substituting 1 for x and $\frac{1}{25}$ for gx in the equation $gx = b^x$ yields $\frac{1}{25} = b^1$, which is equivalent to $\frac{1}{25} = b$. Substituting $\frac{1}{25}$ for b in the equation $gx = b^x$ yields $gx = \frac{1}{25}^x$.

Choice A is incorrect. For this function, $g1$ is equal to -25, not $\frac{1}{25}$.

Choice B is incorrect. For this function, $g1$ is equal to $-\frac{1}{25}$, not $\frac{1}{25}$.

Choice C is incorrect. For this function, $g1$ is equal to 25, not $\frac{1}{25}$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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